

①

Moumita

MATH TEST - 2

Date _____
Page _____

Section - AQuestion - 1

$$|adj A| =$$

Question - 2

$$(c) \quad x = 1.5$$

Question - 3

$$y = \log_e (\cos x)$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{1}{\cos x} \cdot \frac{d}{dx} (\cos x) \\ &= \frac{1}{\cos x} \times (-\sin x) \\ &= -\tan(x) \end{aligned}$$

Question - 4

$$\cos^2 \alpha + \cos^2 \alpha + \cos^2 \alpha = 3\cos^2 \alpha = 1$$

$$\cos \alpha = \pm \frac{1}{\sqrt{3}}$$

$$\text{Direction cosines} = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right)$$

Question - 5

$$P(A \cap B) = P(A) \cdot P(B) = 0.12$$

Question-6

$$R = (-1, 1)$$

Question-7

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

$$A + A^T = \begin{bmatrix} 2 & 5 \\ 5 & 8 \end{bmatrix}$$

Question-8

$$\int e^x (\sin x + \cos x) dx$$

$$= e^x \sin x + C$$

$$a, a_2 + b, b_2 + c, c_2 \geq 0$$

Question-9

$$m \times n = 12$$

$$1 \times 12, 2 \times 6, 3 \times 4, 4 \times 3, 6 \times 2, 12 \times 1$$

Question-10

$$\vec{a} = 2\hat{i} - \hat{j} + 2\hat{k}$$

$$\vec{b} = -\hat{i} + \hat{j} - \hat{k}$$

$$\vec{a} + \vec{b} = (2-1)\hat{i} + (-1+1)\hat{j} + (2-1)\hat{k}$$

$$= \hat{i} + \hat{k}$$

$$|\vec{a} + \vec{b}| = \sqrt{1^2 + 0^2 + 1^2} = \sqrt{2}$$

Unit vector:

$$\frac{\vec{a} + \vec{b}}{|\vec{a} + \vec{b}|} = \frac{\hat{i} + \hat{k}}{\sqrt{2}}$$

Section - B

Question - 11

$$\sin^{-1} \left(\sin \frac{2\pi}{3} \right)$$

$$\sin^{-1} \left(\sin \left(\pi - \frac{\pi}{3} \right) \right)$$

$$= \sin^{-1} \left(\sin \left(\frac{\pi}{3} \right) \right)$$

$$= \sin^{-1} \left(\sin \frac{2\pi}{3} \right)$$

$$= \sin^{-1} \left(\sin \frac{\pi}{3} \right)$$

$$= \frac{\pi}{3}$$

Question - 12

$$A^T = -A$$

$$|A^T| = |-A|$$

$$|A| = |A|$$

(9)

Question - 13

$$\vec{a} = \hat{i} - \hat{j} + 3\hat{k}$$

$$\vec{b} = 2\hat{i} - 7\hat{j} + \hat{k}$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 3 \\ 2 & -7 & 1 \end{vmatrix} = (-1+21)\hat{i} - (1-6)\hat{j} + (-7+2)\hat{k}$$

$$= 20\hat{i} + 5\hat{j} - 5\hat{k}$$

$$\text{Area} = |\vec{a} \times \vec{b}| = \sqrt{20^2 + 5^2 + (-5)^2}$$

$$= \sqrt{400 + 25 + 25}$$

$$= \sqrt{450}$$

$$= 15\sqrt{2} \text{ sq. units.}$$

Question - 14

$$R = \{(a, b) : a \leq b^3\}$$

$$\text{Let } a = 5, b = 2, c = 3$$

$$a \leq b^3 \Rightarrow 5 \leq 2^3$$

$$= 5 \leq 8$$

True. So, $8 \in R$

$$b \leq c^3 \Rightarrow 2 \leq 3^3$$

$$2 \leq 27$$

True. So, $27 \in R$

For transitivity, $(a, c) \in R$

$$a \leq c^3 \Rightarrow 5 \leq 3^3$$

$$5 \leq 27$$

True.

∴ The relation R is transitive.

(5)

Question - 15

$$y = \tan^{-1} \left(\frac{\sin(x)}{1 + \cos(x)} \right)$$

$$\sin(x) = 2 \sin\left(\frac{x}{2}\right) \cos\left(\frac{x}{2}\right)$$

$$1 + \cos(x) = 2 \cos^2\left(\frac{x}{2}\right)$$

$$\Rightarrow \tan^{-1} \left(\tan\left(\frac{x}{2}\right) \right)$$

$$= \left(\frac{x}{2}\right)$$

$$\frac{dy}{dx} = \frac{1}{2}$$

Question - 16

$$S = 36$$

Event A : sum is 6

$$\Rightarrow A = \{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\}$$

$$n(A) = 5$$

Event B : no. 4 appears atleast once

$$A \cap B = \{(2, 4), (4, 2)\}$$

$$n(A \cap B) = 2$$

$$P(A/B) = \frac{n(A \cap B)}{n(B)} = \frac{2}{5}$$

Question - 17

$$\vec{a} = 2\hat{i} - 3\hat{j} + 6\hat{k}$$

$$\vec{b} = \hat{i} + 2\hat{j} + 2\hat{k}$$

$$\vec{a} \cdot \vec{b} = (2)(1) + (-3)(2) + (6)(2)$$

$$= 2 - 6 + 12$$

$$= 8$$

$$|\vec{b}| = \sqrt{1^2 + 2^2 + 2^2} = \sqrt{1 + 4 + 4} = \sqrt{9} = 3$$

$$\text{Scalar projection} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} = \frac{8}{3}$$

Section - C

Question - 18

$$y = x^{\sin x} + (\sin x)^{\cos x}$$

$$\text{Let } y = u + v, \text{ where}$$

$$u = x^{\sin x}$$

$$v = (\sin x)^{\cos x}$$

$$\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$$

$$\text{For } u = x^{\sin x},$$

$$\ln u = \sin x \ln x \Rightarrow \frac{1}{u} \frac{du}{dx}$$

$$= \cos x \ln x + \frac{\sin x}{x}$$

$$\frac{du}{dx} = x^{\sin x} \left(\cos x \ln x + \frac{\sin x}{x} \right)$$

For $y = (\sin x)^{\cos x}$,

$$\ln y = \cos x \ln (\sin x) \Rightarrow \frac{1}{y} \frac{dy}{dx}$$

$$= -\sin x \ln (\sin x) + \cos x \cdot \frac{\cos x}{\sin x}$$

$$= -\sin x \ln (\sin x) + \cot x \cos x$$

$$\frac{dy}{dx} = (\sin x)^{\cos x} (\cos x \cot x - \sin x \ln (\sin x))$$

$$\frac{dy}{dx} = x^{\sin x} \left(\cos x \ln x + \frac{\sin x}{x} \right) + (\sin x)^{\cos x} (\cos x \cot x - \sin x \ln (\sin x))$$

Question-19

$$f(x) = \begin{cases} (\sin 3x)/x, & x < 0 \\ 3, & x = 0 \\ \frac{e^{3x}-1}{x}, & x > 0 \end{cases}$$

LHL

$$\lim_{x \rightarrow 0^-} \frac{\sin 3x}{x} = \lim_{x \rightarrow 0^-} 3 \cdot \frac{\sin 3x}{3x} = 3 \times 1 = 3$$

RHL

$$\lim_{x \rightarrow 0^+} \frac{e^{3x}-1}{x} = \lim_{x \rightarrow 0^+} 3 \cdot \frac{e^{3x}-1}{3x} = 3 \times 1 = 3$$

Value at $x = 0 = 3$

$$f(0) = 3$$

Since $LHL = RHL = f(0) = 3$,

$f(x)$ is continuous at $x = 0$.

8

Question - 20

$$f(x) = 3\sin x + \cos x, \quad x \in [0, 2\pi]$$

Question - 21

5 red, 5 black

Total = 10

Let E_1 be the event of drawing Red on 1st draw,
 E_2 be the event of drawing Black on 1st draw,
 A be drawing Red on 2nd draw

$$P(E_1) = \frac{5}{10} = \frac{1}{2}, \quad P(E_2) = \frac{5}{10} = \frac{1}{2}$$

If Red is drawn first, 2 red balls added.

7 Red + 5 black = 12 total

$$P(A/E_1) = \frac{7}{12}$$

If black is drawn first,

$$P(A/E_2) = 1 - \frac{7}{12} = \frac{5}{12}$$

$$P(A) = P(E_1) P(A/E_1) + P(E_2) P(A/E_2)$$

$$= \frac{1}{2} \times \frac{7}{12} + \frac{1}{2} \times \frac{5}{12}$$

$$= \frac{7+5}{24} = \frac{12}{24}$$

$$\therefore P(A) = \frac{1}{2}$$

Question - 22

$$\tan^{-1} \left(\frac{x-1}{x-2} \right) + \tan^{-1} \left(\frac{x+1}{x+2} \right) = \frac{\pi}{4}$$

$$\tan^{-1} \left(\frac{\frac{x-1}{x-2} + \frac{x+1}{x+2}}{1 - \frac{x-1}{x-2} \times \frac{x+1}{x+2}} \right) = \frac{\pi}{4}$$

$$\frac{(x-1)(x+2) + (x+1)(x-2)}{(x-2)(x+2) - (x-1)(x+1)} = \tan \frac{\pi}{4} = 1$$

$$\frac{x^2 + x - 2 + x^2 - x + 2}{2x^2 - 4} = \frac{x^2 - 4}{2x^2 - 4} = 1$$

$$2x^2 - 4 = 2x^2 - 4$$

$$x^2 = \frac{-1}{2}$$

$$x = \sqrt{\frac{-1}{2}}$$

Question - 23

If $y = (\sin^{-1} x)^2$

$$(1-x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} - 2 = 0$$

$$\frac{dy}{dx} = 2 \sin^{-1}(x) \cdot \frac{1}{\sqrt{1-x^2}}$$

$$\sqrt{1-x^2} \frac{dy}{dx} = 2 \sin^{-1}(x)$$

$$(1-x^2) \left(\frac{dy}{dx} \right)^2 = 4 [\sin^{-1}(x)]^2 = 4y$$

Question - 24

$$\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = abc \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right)$$

$$\Delta = abc \begin{vmatrix} \frac{1}{a}+1 & 1 & 1 \\ 1 & \frac{1}{b}+1 & 1 \\ 1 & 1 & \frac{1}{c}+1 \end{vmatrix}$$

$$R_1 \rightarrow R_1 + R_2 + R_3$$

$$\Delta = abc \begin{vmatrix} 1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c} & 1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c} & 1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c} \\ \frac{1}{b} & \frac{1}{b}+1 & \frac{1}{b} \\ \frac{1}{c} & \frac{1}{c} & \frac{1}{c}+1 \end{vmatrix}$$

$$\Delta = abc \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \begin{vmatrix} 1 & 1 & 1 \\ \frac{1}{b} & \frac{1}{b}+1 & \frac{1}{b} \\ \frac{1}{c} & \frac{1}{c} & \frac{1}{c}+1 \end{vmatrix}$$

$$C_2 \rightarrow C_2 - C_1, \quad C_3 \rightarrow C_3 - C_1$$

$$\Delta = abc \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \begin{vmatrix} 1 & 0 & 0 \\ \frac{1}{b} & 1 & 0 \\ \frac{1}{c} & 0 & 1 \end{vmatrix}$$

$$= abc \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) (1 \cdot (1 \cdot 1 - 0))$$

~~Ans~~

$$= abc \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \cdot 1$$

Section - D

Question - 25

~~$$2x + 3y + 3z = 5$$~~
~~$$x - 2y + z = -4$$~~
~~$$3x - y - 2z = 3$$~~

$$2x + 3y + 3z = 5$$

$$x - 2y + z = -4$$

$$3x - y - 2z = 3$$

$$\begin{bmatrix} 2 & 3 & 3 \\ 1 & -2 & 1 \\ 3 & -1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ -4 \\ 3 \end{bmatrix}$$

$$AB = X$$

$$B = A^{-1}X$$

$$\Delta = 2(4+1) - 3(-2-3) + 3(-1+6)$$

$$= 2 \times 5 - 3 \times -5 + 3 \times 5$$

$$= 10 + 15 + 15$$

$$|A| = 40$$

$$\text{adj } A = \begin{bmatrix} 5 & 5 & 5 \\ 3 & -13 & 11 \\ 9 & 1 & -7 \end{bmatrix}^T = \begin{bmatrix} 5 & 3 & 9 \\ 5 & -13 & 1 \\ 5 & 11 & -7 \end{bmatrix}$$

$$A^{-1} = \frac{\text{adj } A}{|A|} = \frac{1}{40} \begin{bmatrix} 5 & 3 & 9 \\ 5 & -13 & 1 \\ 5 & 11 & -7 \end{bmatrix}$$

$$X = \frac{1}{40} \begin{bmatrix} 5 & 3 & 9 \\ 5 & -13 & 1 \\ 5 & 11 & -7 \end{bmatrix} \begin{bmatrix} 5 \\ -4 \\ 3 \end{bmatrix}$$

$$= \frac{1}{40} \begin{bmatrix} 25 - 12 + 27 \\ 25 - 52 + 3 \\ 25 - 44 - 21 \end{bmatrix}$$

12

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{40} \begin{bmatrix} 40 \\ 80 \\ -40 \end{bmatrix}$$

$$\therefore \boxed{x=1, y=2, z=-1}$$

Question - 26

$R = \{(a, b) : |a-b| \text{ is a multiple of } 4\}$

Reflexive:

For any $a \in A$,
 $|a-a| = 0 = 0 \times 4$ which is
 a multiple of 4. So $(a, a) \in R$

Symmetric: If $(a, b) \in R$
 $|a-b| = 4k$

$$|b-a| = |-(a-b)| = 4k$$

$(b, a) \in R$

Transitive: If $(a, b) \in R$ and $(b, c) \in R$,
 then $a-b = 4k_1$ & $b-c = 4k_2$
 $\therefore R$ is an equivalence.
 ~~$a \sim b$~~

$$\{1, 5, 9\}$$

27) Let E_1, E_2, E_3 be events of work done by A, B, C

$$P(E_1) = 0.5 \quad P(E_2) = 0.3 \quad P(E_3) = 0.2$$

Let D be event of defective items

$$P(D/E_1) = 0.01$$

$$P(D/E_2) = 0.05$$

$$P(D/E_3) = 0.07$$

$$P(E_1/D) = \frac{P(E_1) P(D/E_1)}{P(E_1) P(D/E_1) + P(E_2) P(D/E_2) + P(E_3) P(D/E_3)}$$

$$= \frac{1 \times 0.01}{5}$$

$$0.5 \times 0.01 + 0.05 \times 0.3 + 0.2 \times 0.07$$

~~5/12~~

$$= 0.005$$

$$0.005 + 0.015 + 0.014$$

$$= 0.005$$

$$0.034$$

$$P(E_1/D) = \frac{5}{34} \approx 0.11$$